JOSEPH WU

SENIOR SOFTWARE ENGINEER | AI ENGINEERING

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PROFESSIONAL SUMMARY

Senior Software Engineer and hands-on Engineering Manager with 14 years of experience designing, developing, and maintaining production software across financial technology, backend platforms, and AI-enabled applications. Strong background in Python, Java, Node.js, distributed services, REST APIs, and cloud-based systems, with an emphasis on reliability, maintainability, and operational resilience.

Currently leads AI platform engineering work at Arbitexia, combining architecture and implementation with code reviews, technical guidance, and mentorship. Builds LLM-powered applications, retrieval-augmented generation (RAG) pipelines, semantic retrieval, document-processing services, and agent workflows using tools such as FastAPI, LangChain, LangGraph, and vector search. Focuses on grounded responses, validation, evaluation, and human approval for consequential actions.

Previously developed financial technology services at Wealthsimple and Wave Financial, including asynchronous processing, data integrations, PostgreSQL optimization, and internal tools. Works closely with product, security, and infrastructure partners from system design through deployment and production troubleshooting, balancing delivery speed with code quality and long-term maintainability.

TECHNICAL EXPERTISE

Backend & Languages: Python, Java, Node.js, TypeScript, JavaScript, FastAPI, Django, Flask, REST APIs, SQL

AI & Data: LLM APIs, RAG, LangChain, LangGraph, embeddings, pgvector, semantic retrieval, agent workflows, prompt evaluation, PostgreSQL, Prisma, Redis

Cloud & Delivery: AWS, Docker, Kubernetes, Terraform, CI/CD, asynchronous processing, observability, production operations

Frontend: React, Next.js, HTML, CSS, Tailwind CSS, responsive UI, accessibility

PROFESSIONAL EXPERIENCE

ARBITEXIA | Engineering Manager, AI Platforms

Jan 2025 – Present | Remote from Toronto, Canada

- Lead hands-on development of Python-based AI applications, spanning service architecture, implementation, code reviews, and delivery planning.
- Design FastAPI services and RAG pipelines for document ingestion, semantic retrieval, and source-grounded responses.
- Prototype LLM agent workflows with tool integrations, validation checks, audit trails, and human approval for consequential actions.
- Guide engineering practices for testing, prompt and retrieval evaluation, deployment, and production troubleshooting.
- Work with stakeholders to define achievable releases and balance product requirements with maintainable architecture.

PROFESSIONAL EXPERIENCE | CONTINUED

WEALTHSIMPLE | Senior Software Engineer

Apr 2019 – Dec 2024 | Toronto, Ontario, Canada

- Built backend services and APIs supporting financial data workflows, integrations, and internal operations.
- Designed asynchronous processing with retries, idempotency, and observability to strengthen service resilience.
- Optimized PostgreSQL queries and caching paths; improved automated tests, release checks, and incident diagnostics.
- Collaborated with product, security, and infrastructure teams on system design and production rollouts; mentored engineers through reviews and pairing.
- Developed internal React and TypeScript interfaces for workflow visibility and operational investigations.

WAVE FINANCIAL | Software Engineer

Jul 2015 – Mar 2019 | Toronto, Ontario, Canada

- Developed and maintained Python/Django services and REST APIs supporting small-business financial workflows.
- Implemented background jobs, third-party API integrations, and data synchronization with clear failure handling.
- Improved SQL access patterns, automated test coverage, and logging in collaboration with product and frontend teams.

WAVE FINANCIAL | Junior Software Engineer

Jul 2012 – Jun 2015 | Toronto, Ontario, Canada

- Delivered application features using Python, Django, JavaScript, and relational databases with guidance from senior engineers.
- Investigated defects, added validation and tests, and supported deployments and third-party integrations.

SELECTED ENGINEERING WORK

AI Knowledge Assistant — Retrieval-Augmented Generation Designed a RAG architecture covering document ingestion, chunking, vector search, retrieval APIs, and source-grounded answers. Connected retrieval results to generated responses to keep outputs tied to supporting documents.

Workflow Automation — LLM Tool Calling Prototyped a tool-calling workflow with deterministic validation checks and explicit approval steps before consequential actions, emphasizing predictable behavior and human oversight.

EDUCATION

University of Waterloo | Bachelor of Computer Science
2008 – 2012 | Waterloo, Ontario, Canada